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Module 6 Introduction

Lesson 1: Bias, Variance and Model Complexity

Reading: Bias, Variance and Model Complexity Readings
1h 15m

Video: Bias, Variance and Model Complexity
10 min

Video: The Bias-Variance Decomposition
8 min

Practice Assignment: Bias, Variance and Model Complexity
Submitted

Practice Assignment: Bias, Variance and Model Complexity Quiz 2
Submitted

Lab: Coding Example
1h

Lesson 2: Bayesian Approach and BIC

Lesson 3: Vapnik-Chernovenkis (VC) Dimension

Lesson 4: Cross Validation

Lesson 5: Bootstrapping

Module 6 Summative Assessment

Module 6 Summary

Your grade: 100%

Next item

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Bias, Variance and Model Complexity Quiz 2

Instructions

Review Learning Objectives

1. In the context of kNN, what is meant by 'feature scaling'? **1 / 1 point**

- (●) Adjusting the range of different features to make them comparable.
- ( ) Selecting the most important features for the model.
- ( ) Transforming categorical features into numerical ones.
- ( ) Reducing the number of features used in the model.

Correct
Correct! Feature scaling in kNN ensures that all features contribute equally to the distance calculation.

2. Why is the choice of distance metric important in kNN? **1 / 1 point**

- ( ) It dictates the type of problem kNN can solve.
- ( ) It determines the number of neighbors to be considered.
- ( ) It is the primary factor in determining the model's accuracy.
- (●) Different metrics can influence how the 'nearest neighbors' are determined.

Correct
Correct! The choice of distance metric affects the calculation of nearest neighbors in kNN.

3. What is a common challenge when using kNN for large datasets? **1 / 1 point**

- ( ) It automatically reduces the dimensionality of the data.
- ( ) It tends to overfit regardless of the dataset size.
- (●) The algorithm can become computationally intensive as it requires distance calculations for each query.
- ( ) It requires a fixed number of features for all data points.

Correct
Correct! kNN's need to compute distances for each query can be demanding for large datasets.

4. In kernel smoothing, what is the effect of using a larger bandwidth? **1 / 1 point**

- (●) It results in a smoother estimate but may oversmooth important features in the data.
- ( ) It increases the sensitivity of the smoother to outliers.
- ( ) It reduces the computational complexity of the smoothing process.
- ( ) It creates a more complex model with more peaks and valleys.

Correct
Correct! Larger bandwidths in kernel smoothing produce smoother curves but can oversmooth.

5. What is a potential drawback of local regression? **1 / 1 point**

- ( ) It is computationally less intensive than global regression models.
- ( ) It always results in linear relationships between variables.
- ( ) It requires large amounts of data to perform any fitting.
- (●) It can be sensitive to the choice of bandwidth, affecting the smoothness of the fit.

Correct
Correct! The choice of bandwidth is crucial and can greatly affect the outcome of local regression.

6. In which scenario would local regression be particularly useful? **1 / 1 point**

- (●) When the data exhibits varying patterns or trends in different regions.
- ( ) When the primary goal is to reduce the dimensionality of the dataset.
- ( ) When the data is uniformly distributed with a clear global trend.
- ( ) When all variables in the dataset are categorical.

Correct
Correct! Local regression is ideal for data with non-uniform patterns across different regions.